



AI-Powered Explainable Linux Security Assistant for Kernel-Level Intrusion & Behavioral Threat Detection



TEAM NAME: ZEPHYR



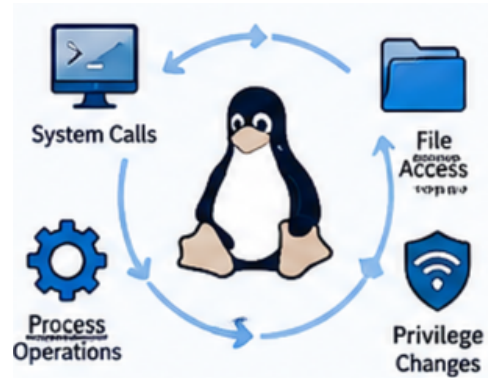
INSTITUTE: BANNARI AMMAN INSTITUTE OF TECHNOLOGY

PROBLEM STATEMENT

Need for an Intelligent, Real-time Linux Security Assistant for Kernel-level Threat Detection

THE PROBLEM

1. CONTINUOUS KERNEL ACTIVITY



High volume of system calls, process operations, file access, networking, and privilege changes are difficult to monitor effectively.

2. RELIANCE ON FIXED SIGNATURES & RULES



Existing security tools rely on fixed signatures and rules, making them less effective against unknown and evolving threats.

3. POST-EVENT LOG ANALYSIS



Security tools often depend on post-event log analysis, leading to delayed detection and slower response.

4. STEALTHY MALICIOUS BEHAVIOR



Malicious processes can exhibit abnormal behavior before visible symptoms appear, increasing the risk of compromise.

5. NEED FOR INTELLIGENT, REAL-TIME ASSISTANT



There is a need for a system that learns normal behavior, detects deviations, explains anomalies, and recommends or performs corrective actions in real time.

WHAT AN INTELLIGENT LINUX SECURITY ASSISTANT SHOULD DO



1. COLLECT REAL-TIME KERNEL ACTIVITY



2. LEARN NORMAL BEHAVIOR



3. DETECT ANOMALIES IN REAL TIME



4. EXPLAIN THE REASON



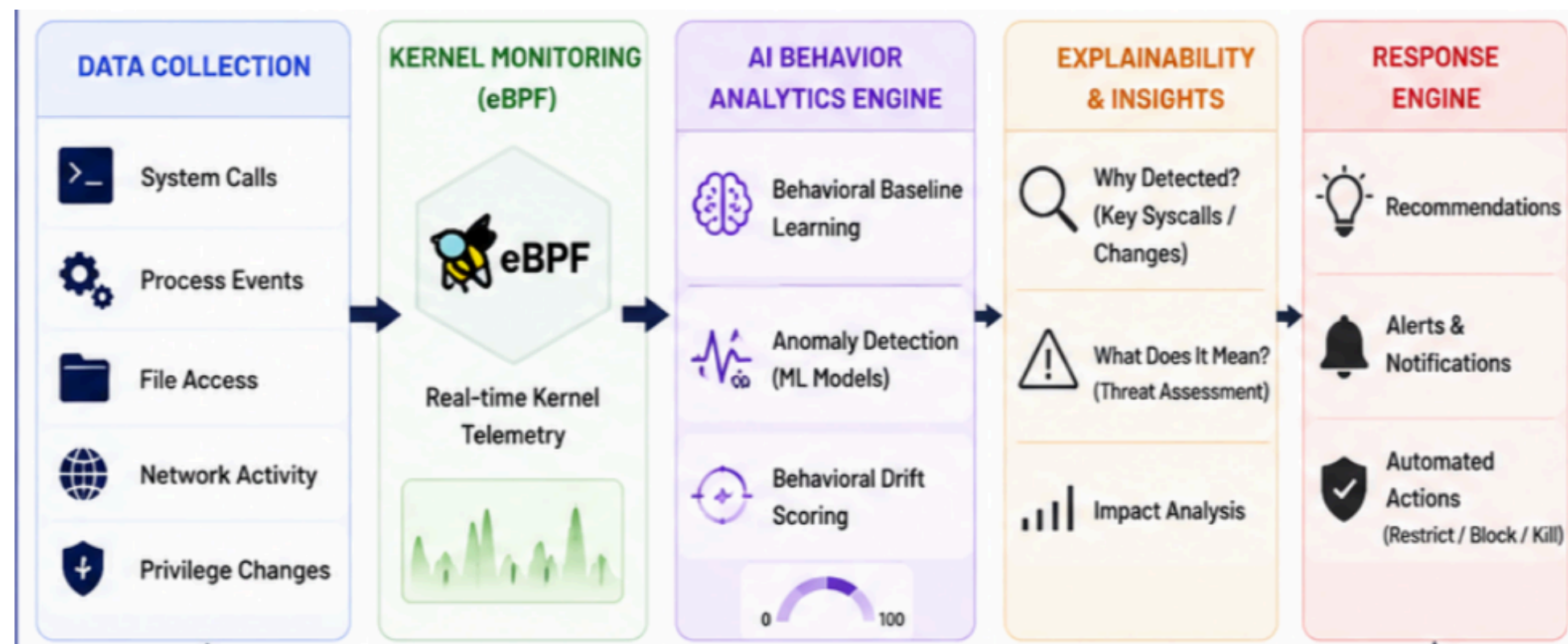
5. RECOMMEND OR PERFORM ACTIONS

PROPOSED SOLUTION

AI-Powered Explainable Linux Security Assistant for Kernel-Level Intrusion & Behavioral Threat Detection

OUR PROPOSED SOLUTION

We propose an AI-Powered Explainable Linux Security Assistant that monitors kernel-level behavior in real time, learns normal process patterns detects anomalies, explains the reason, and recommends or performs appropriate corrective actions.



KEY CAPABILITIES

- 01 Real-time Kernel Monitoring**
Captures system calls, process, file, network, and privilege activities using eBPF.
- 02 Behavioral Baseline Learning**
Learns normal behavior patterns for each process over time.
- 03 Anomaly Detection**
Detects deviations from normal behavior using AI/ML models.
- 04 Explainable AI**
Highlights the exact reasons for anomalies (e.g., unusual syscalls, privilege changes).
- 05 Threat Assessment**
Classifies threats by severity and estimates potential impact.
- 06 Action Recommendations**
Suggests appropriate actions (alert, restrict, block, isolate, or kill).
- 07 Automated Response**
Performs safe, policy-based actions for high-severity threats.

BENEFITS

-  **DETECTS THREAST EARLY**
-  **PROTECTS SYSTEM CONTINOUSLY**
-  **EXPLAINS ANOMALIES**
-  **ENABLES SMART RESPONSE**
-  **ENHANCE SECURITY &RELIABILITY**

OBJECTIVE AND APPROACH

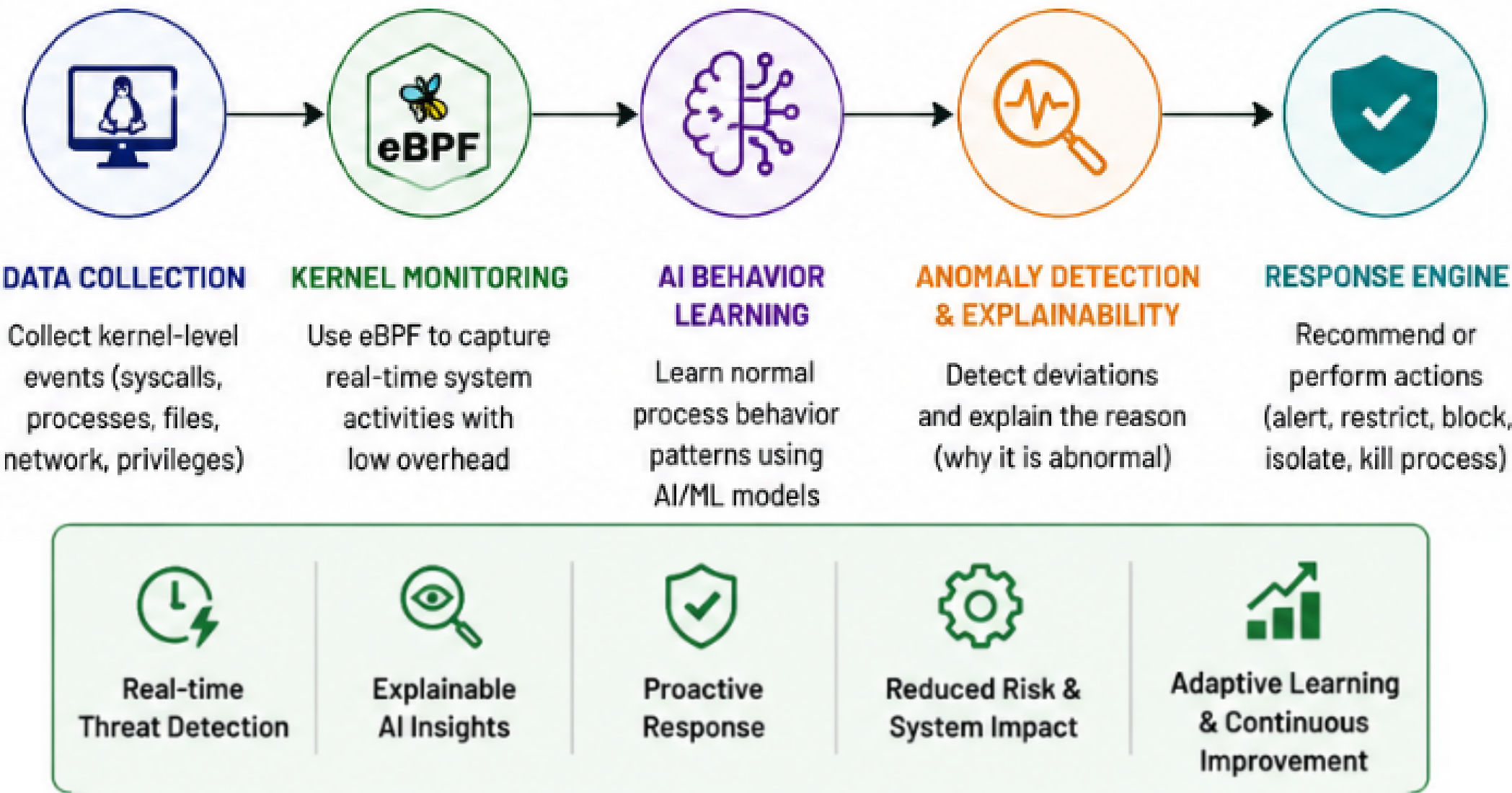
AI-Powered Explainable Linux Security Assistant for Kernel-Level Intrusion & Behavioral Threat Detection

OBJECTIVE

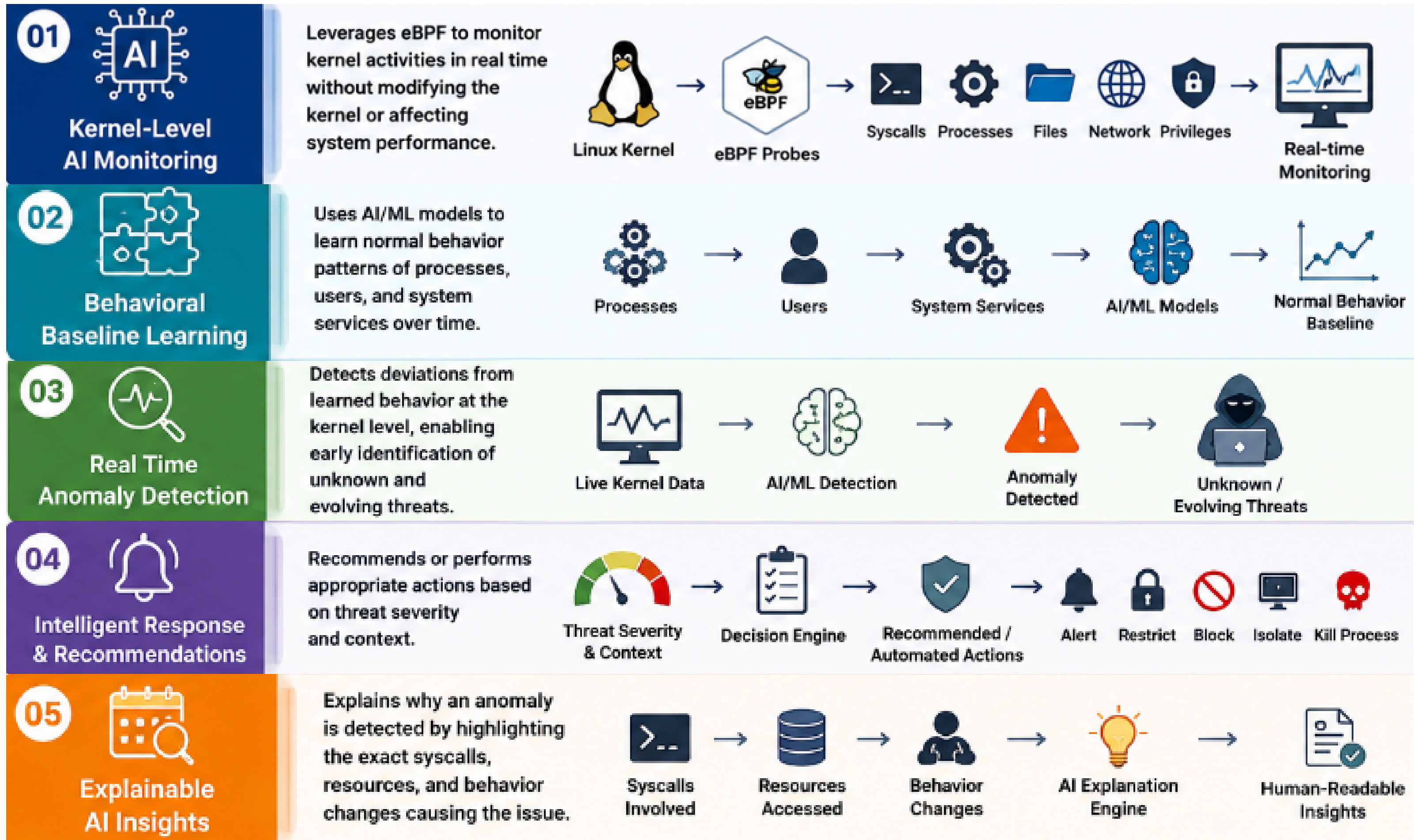
-  To develop an AI-powered Linux security assistant that learns normal process behavior, detects anomalies, explains the reason, and recommends or performs corrective actions in real time.
-  Enable real-time monitoring of kernel-level activities including system calls, processes, files, networks, and privilege changes.
-  Learn and model normal process behavior using AI/ML for accurate baseline creation.
-  Detect behavioral deviations and unknown threats at the kernel level.
-  Provide explainable insights by identifying the exact cause of anomalies.
-  Recommend or perform appropriate actions to mitigate threats and secure the system.

APPROACH

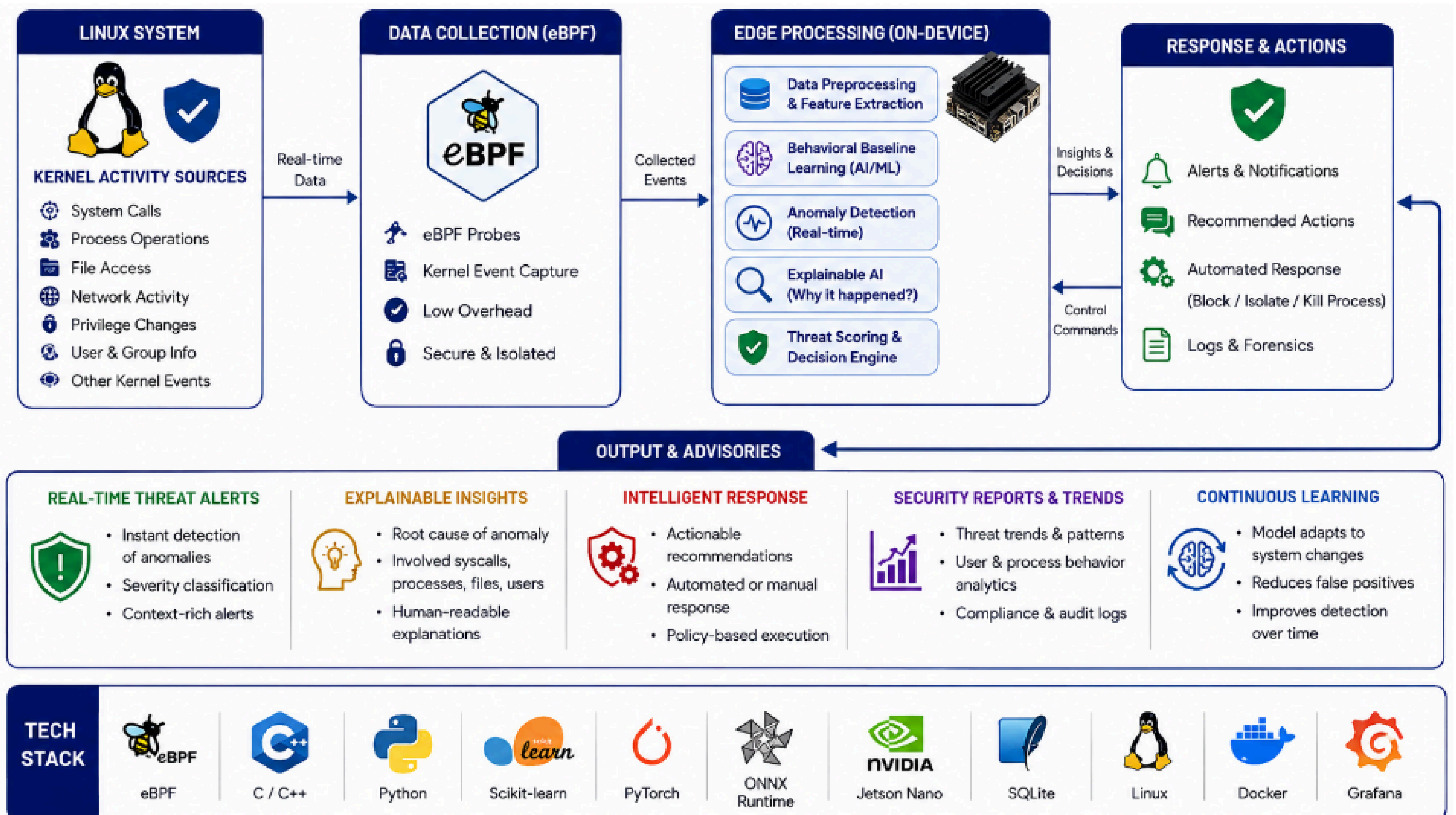
Our approach is to build an intelligent Linux security assistant that monitors kernel activities in real time, learns normal behavior using AI, detects anomalies, explains the reason, and recommends or performs corrective actions automatically.



NOVELTY



TECHANICAL APPROACH



EXISTING TECHNIQUES

WHAT IS CURRENTLY USED?



Signature / Rule-Based IDS

Detects known threats using predefined signatures or rules.



Log-Based Analysis

Relies on system logs analyzed after an event to identify suspicious activities.



Host-Based Firewalls

Controls inbound and outbound traffic based on predefined rules and policies.



File Integrity Monitoring (FIM)

Monitors critical files and alerts on unauthorized changes.



Access Control & Permissions

Uses UNIX permissions, ACLs and SELinux/AppArmor policies to restrict access.



Manual Security Audits

Security checks performed manually using commands and scripts at intervals.

LIMITATIONS



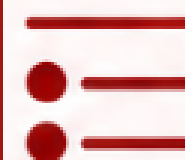
Reactive, Not Predictive

Threats are detected only after they occur, leading to potential damage.



No Early Anomaly Detection

Abnormal behavior at kernel level often goes unnoticed until it becomes severe.



Signature & Rule Dependence

Ineffective against unknown, zero-day and evolving threats.



Post-Event Analysis

Log-based approaches cause delayed response and increased risk.



Cloud Dependency

Requires network connectivity, raising latency, privacy and operational risks.



No Intelligent Response

Most solutions only alert; they do not recommend or perform corrective actions automatically.

BENEFITS AND LIMITATIONS

BENEFITS



Proactive Threat Detection

Detects unknown and evolving threats in real time at the kernel level.



AI-Powered Intelligence

Learns normal behavior of processes and users to accurately identify anomalies.



Explainable Insights

Explains why an anomaly occurred by highlighting the exact cause, syscalls and behavior changes.



Intelligent Response

Recommends or performs appropriate actions (alert, restrict, isolate, kill) based on threat severity.



Improved System Security

Strengthens overall security posture without modifying the kernel or affecting system stability.

LIMITATIONS



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